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clusterExplorer: an R/Shiny app for single-cell RNA-seq cluster visualizations

Background

Single-cell RNA sequencing (scRNA-seq) is a powerful biological technique that offers valuable insight into cellular processes and related structures, but the comparison between different datasets is no trivial task. UMAP- [1] or t-SNE-based [2] clusterings can help to structure the scRNA data and allow insight into different cell populations, but different resolutions for these clusterings provide different benefits. An analysis of more detailed clusters from a subgroup of samples can reveal specific information, while a general overview may be more desirable for a standard analysis of larger datasets. In certain cases however, the location of these more specific structures may still be biologically meaningful with regards to the larger dataset, i.e. for the analysis of tumor cell populations or pairs of primary tumor and relapse data. The R/Shiny [3] app clusterExplorer offers visualization routines for scRNA-seq clusters in datasets with predefined subsets, e.g. subpopulations of tumor types, integrated datasets containing original and published scRNA-seq data, or datasets with different biological conditions like tumor and relapse data. For each predefined subset of a chosen dataset, the cells of any original cluster can be traced, compared, and classified within the clustering of the full set.

Methods

clusterExplorer uses a standardized R-based [4] preprocessing pipeline for each dataset. This pipeline consists of an alignment step with the CellRanger algorithm [5] and the Seurat workflow [6], and is used to obtain cluster information for each cell. Further data preprocessing with R is conducted separately for each dataset and subset to provide the basic statistics for cluster-based scoring schemes and comparisons. clusterExplorer subsequently allows the user to select one of Seurat's UMAP- or t-SNE-based clusters from the chosen dataset or subset for parallel visualization in the second dataset, provided that this set contains subsets of the chosen cells. For the resulting cell cluster projections, the cluster identities are compared between the datasets, and the highest overlaps in the other dataset's clustering are provided (Figure 1). In addition, clusterExplorer offers a convex hull representation of each cluster, in which transparent overlays for chosen quantiles of the cells allow to assess the general cluster structure. Quantiles of the data (0.25, 0.5, 0.75 and 1.0) are visualized; for each of the resulting subsets, the median distance between points, the standard deviation, and the area of the resulting convex hull are calculated; an example is provided in Figure 2. The structure of the original clusters and the structure of the traced or "projected" cells are classified based on their median distance and area: more "compact" clusters are defined as clusters with a relatively low median distance between point pairs and more modest increases in area, while "scattered" clusters show a greater distance between points and larger relative areas for the 0.75 or 1.0 quantiles. The comparison between both subset and full dataset allows to identify conserved or split structures, depending on the overlaps of original cluster cells with the second set's clustering. If one matching cluster exists for $> 75\%$ of the original cluster's cells in the projected dataset, we call this cluster "conserved"; if the two best-matching

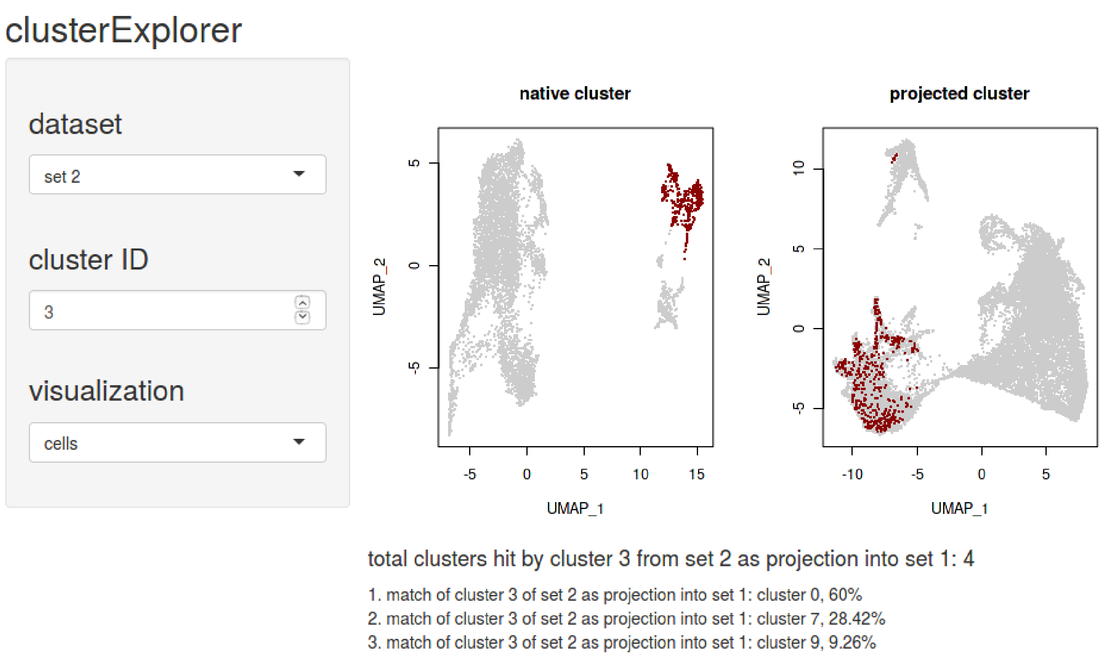

Figure 1: Example output of clusterExplorer's cell-based visualizations for ependymoma data from Gillen et al. [7]. Cells that form a more compact cluster in a subset from Gillen et al.'s data (left) are distributed over multiple clusters in the projection, however more than 75% of these cells overlap with two clusters in the full dataset (right). Thus, the cluster's structure is defined as split when comparing the ependymoma subset and full dataset.

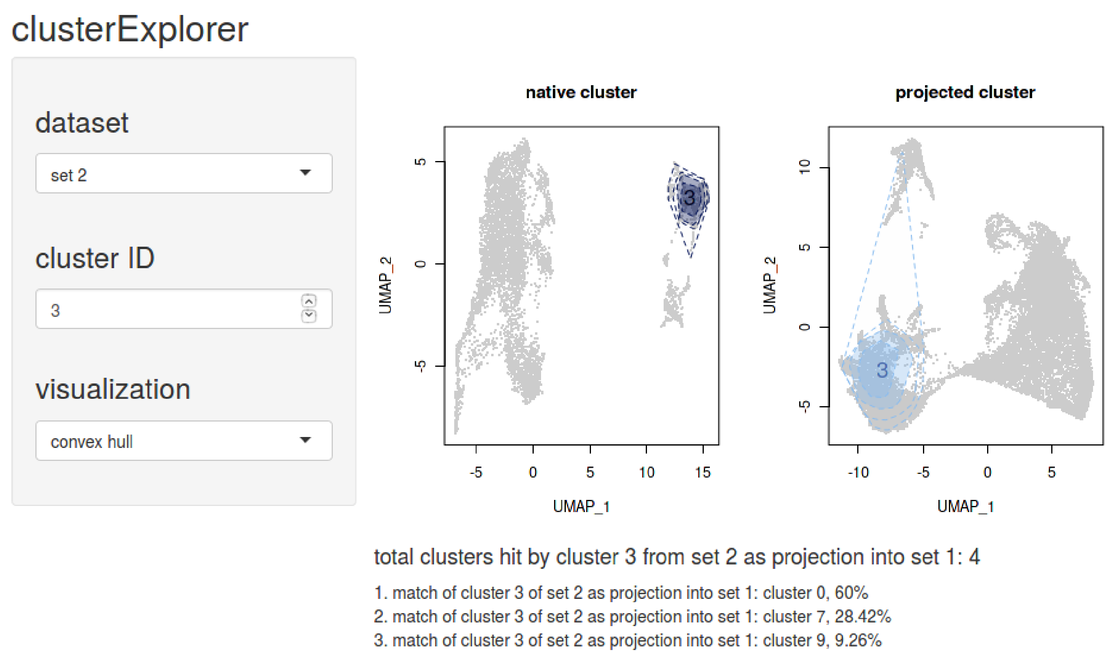

Figure 2: Convex hull visualizations for the chosen test cluster: For each chosen quantile (0.25, 0.5, 0.75, 1.0), a convex hull is drawn, with shades of blue colour depicting more compact clusters, and red lines visualizing scattered clusters.

clusters exceed 75%, the cluster is defined as "split". Otherwise, notable parts of the original cluster structure are distributed over several clusters in the second dataset.

Results and discussion

For the ependymoma test data of Gillen et al., two of the 11 original clusters were classified as "compact" and two as "scattered", the remaining clusters were intermediate (Figure 3). Five cluster structures were conserved between the two clusterings, four clusters were split into two new clusters for the projected data; the remaining three original clusters were distributed.

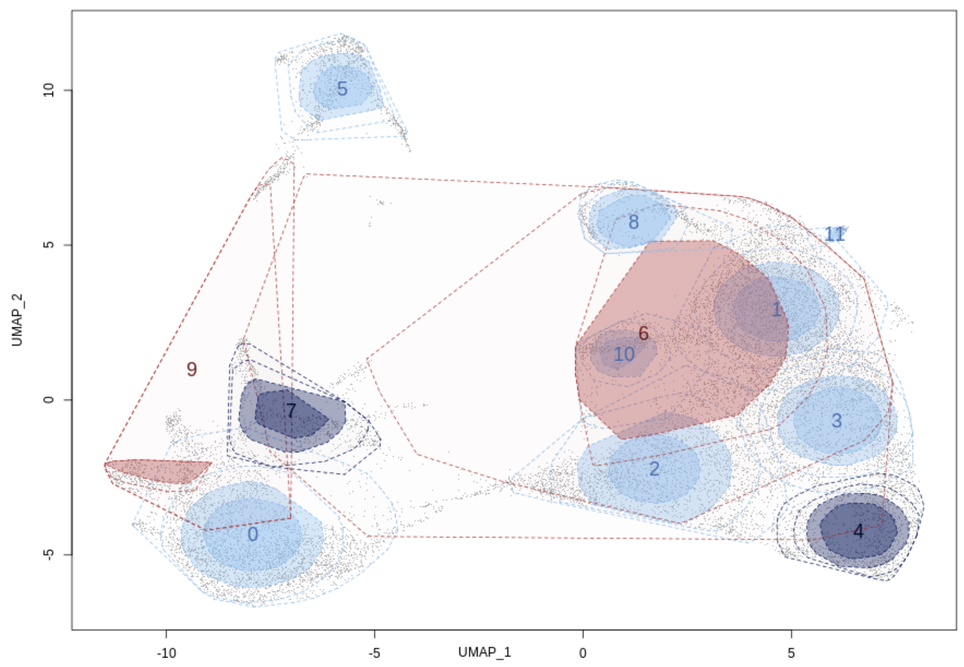

Figure 3: Combined visualization of convex hulls per cluster for the full test set. Semi-transparent overlays show the quantiles 0.25 and 0.5 for the more compact or intermediate clusters, and the 0.25 quantile for scattered clusters.

Identification of compact and scattered clusters and conserved or split cluster structures between clusterings may help to find interesting cell populations or problematic parts in clusterings. The current classification of clusters is sensitive to larger changes in distance- or area-based distributions, however, and may need to be adjusted for other datasets.

References

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